

Exp 1- Configure an IP Network

What are the ARP packets and what are they used for?

Quais são

```
Request-> Who has 172.16.120.1? Tell 172.16.120.254
Reply->    172.16.120.254 is at ec:75:0c:c2:3c:ac
Request-> Who has 172.16.120.254? Tell 172.16.120.1
Reply->    172.16.120.1 is at ec:75:0c:c2:3c:75
```

Para que servem

Estes pacotes servem para fazer o mapeamento entre os endereços IP e os endereços MAC dentro de uma rede local

What are the MAC and IP addresses of ARP packets and why?

Quais são

```
Tux3:
- IP: 172.16.120.1
- MAC: ec:75:0c:c2:3c:75
```

```
Tux4:
- IP: 172.16.120.254
- MAC: ec:75:0c:c2:3c:ac
```

Porquê

Porque são os que aparecem no ficheiro de log do WireShark e porque coincidem com os endereços registados ao fazer o setup da rede

What packets does the ping command generate?

Pacotes ICMP. Pacotes que ajudam a identificar problemas na rede

What are the MAC and IP addresses of the ping packets?

```
Tux3:
- IP: 172.16.120.1
- MAC: ec:75:0c:c2:3c:75
```

```
Tux4:
- IP: 172.16.120.254
- MAC: ec:75:0c:c2:3c:ac
```

How to determine if a receiving Ethernet frame is ARP, IP, ICMP?

Na coluna Protocol, no WireShark, está indicado o tipo de pacote utilizado

How to determine the length of a receiving frame?

Na coluna Length, no WireShark, está indicado o tamanho do pacote

What is the loopback interface and why is it important?

O que é

É uma interface de rede virtual reservada que permite ao computador comunicar consigo próprio.

Porque é importante

Com isto é possível que diferentes serviços ou aplicações a correr no mesmo computador comuniquem entre si utilizando protocolos de redes standart

Exp 2 - Implement two bridges in a switch

How to configure bridge120?

No MTKterm:

- /interface bridge add name=bridgeY0
- /interface bridge port remove [find interface =ether1]
- /interface bridge port add bridge=bridgeY0 interface=ether1.

How many broadcast domains are there?

2

How can you conclude it from the logs?

Ao fazer broadcast a partir do tux3 apenas o tux 4 ouve
Ao fazer broadcast a partir do tux2 mais nenhum tux ouve

Exp 3 - Configure a Router in Linux

What routes are there in the tuxes? What are their meaning?

tux2:

- Destination: 172.16.120.0; Gateway: 172.16.121.253

. 0 tux2 consegue aceder à rede do tux3 (172.16.120.0) através da gateway 172.16.121.253 (ip do tux4, que estabelece a comunicação entre as duas redes)

tux3:

- Destination: 172.16.121.0; Gateway: 172.16.120.254
. 0 tux3 consegue aceder à rede do tux2 (172.16.121.0) através da gateway 172.16.120.254 (ip do tux4, que estabelece a comunicação entre as duas redes)

What information does an entry of the forwarding table contain?

A tabela contém as rotas existente onde, para cada uma, podemos ver o Destino, a Gateway de acesso ao Destino, a Máscara da rede de destino, as Flags associada à rota, e as Interface que a rota usa.

What ARP messages, and associated MAC addresses, are observed and why?

Mensagens e endereços MAC

tux3:

16 - Who has 172.16.120.1? Tell 172.16.120.254
17 - 172.16.120.1 is at ec:75:0c:c2:3c:ac
18 - Who has 172.16.120.254? Tell 172.16.120.1
19 - 172.16.120.254 is at ec:75:0c:c2:3c:75

72 - Who has 172.16.120.1? Tell 172.16.120.254
73 - 172.16.120.1 is at ec:75:0c:c2:3c:ac

tux4 if_e1:

25 - Who has 172.16.120.254? Tell 172.16.120.1
26 - 172.16.120.254 is at ec:75:0c:c2:3c:75
27 - Who has 12.16.120.1? Tell 172.16.120.254
28 - 172.16.120.1 is at ec:75:0c:c2:3c:ac

tux4 if_e2:

31 - Who has 172.16.121.253? Tell 172.16.121.1
32 - 172.16.121.253 is at ec:75:0c:c2:10:6b
33 - Who has 12.16.121.1? Tell 172.16.121.253
34 - 172.16.121.1 is at ec:75:0c:c2:31:73

Why

As mensagens ARP permitem mapear a que endereço MAC está associado cada endereço IP

What ICMP packets are observed and why?

Vários pacotes "Echo (ping) request" e "Echo (ping) reply"

Inicialmente o ping do tux3 com destino ao tux2 aparece como uma resposta do tux4, após serem limpas as tabelas arp, as respostas ao ping do tux3 são registadas como tendo origem no IP associado ao tux2; Isto acontece porque ao ser redirecionada a mensagem de ping recebe o IP do tux4, que a reenvia. Depois de serem apagadas as tabelas arp, os campos de Destino e Origem das mensagens já aparecem com os valores corretos de IP do tux2 e do tux3.

What are the IP and MAC addresses associated to ICMP packets and why?

```
tux3 if_e1 172.16.120.1 - ec:75:0c:c2:3c:ac
tux4 if_e1 172.16.120.254 - ec:75:0c:c2:3c:75
tux4 if_e2 172.16.121.253 - ec:75:0c:c2:10:6b
tux2 if_e1 172.16.121.1 - ec:75:0c:c2:31:73
```